

# 31338 Network Servers

## Lab 5b: Superuser Powers and System Notifications

### Aims

1. Learn how to safely use root/administrator access
2. Notify users about system issues or company computer usage policies

### Task 1: Executing Root Commands via sudo

In CentOS you simply have logged on as root to execute privileged commands. This is extremely dangerous since any command can be executed, and there is no real audit of who used the root account.

The best practice is to make use of the *sudo* command to execute specific commands.

From root, see the */etc/sudoers* file to inspect the system permissions. However, we do not directly edit this file using a text editor because the invalid format or incorrect settings lead to dangerous consequences. Instead, we use the command *visudo* to change the properties.

By default, a standard user account cannot execute the many commands like *fdisk*. Can you switch to the user peter (either ssh or su) and execute *fdisk*?

The format of *sudoers* file defines users and the set of commands that the user may run. The format is like

```
<user_name> <MACHINE>=<COMMANDS>
```

Look through the file that *visudo* displays. What commented-out user permissions are there? These are standard permissions a system administrator might wish to grant.

The *fdisk* executable is in */sbin* directory. Add the following row to *sudoers*

```
peter  ALL=/sbin/fdisk
```

Save and close through *visudo*. Switch to the user peter. How can you execute *fdisk -l* with this user account rather than root?

### Task 2: Executing Administrator Programs in Windows Server

Windows UAC (User Account Control) controls whether the commands you execute that need the administrator privilege and will prompt you whether to do so. You can usually execute a command with administrator *Run as Administrator* in the GUI.

## Task 3: The Message of the Day (MOTD)

When a user logs in the CentOS from a terminal, a system administrator can set up a welcome message. Edit the file `/etc/motd` and type a message. Then switch to any user using `ssh`. Can you see the message you typed in?

## Task 4: GNOME Login Screen Notifications (GUI)

CentOS GUI uses `gdm` (Gnome Display Manager) to show the messages. You need to do the following steps to set it up.

1. Create a new file `/etc/dconf/profile/gdm`, then add the following rows:  

```
user-db:user
system-db:gdm
```
2. Create a configuration file `/etc/dconf/db/gdm.d/01-banner-message`, then add the followings:  

```
[org/gnome/login-screen]
banner-message-enable=true
banner-message-text="Welcome to FEIT"
```
3. Update the dconf database by running  

```
dconf update
```
4. Restart the `gdm` service by running  

```
systemctl restart gdm
```

Now logout the current user and re-login through GUI. Can you see the welcome message?

## Task 5: Windows Server Login Messages

On Windows Server, you can also set a welcome message or company security policy message when a user logs in.

1. Go to *Server Manager* -> *Tools* -> *Local Security Policy*.
2. From the list of security settings on the left pane, navigate to *Local Policies* -> *Security Options*. Scroll down and locate the two entries that start with "Interactive log on".
3. This activity requires us to edit two of them. Double click on the name and enter your text messages.
4. Open a terminal of PowerShell and run  

```
gpupdate /force
```

  
to update the system policies.

Log out and log back in again. Your message should be shown.